

(1001CJA621621005)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

UNIT TEST

28-03-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	B,D	A,C	A,B,C,D	A,B,D	A,B,D	A	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	47.77 to 47.78	25.00	2.20	1.84	10.20	3.00	4.00	12.90 to 13.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C	A,B,D	A,B	B,D	A,B,C	D	C	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	0.03	1.33	5.00	30.00	83.00	3.00	375.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C	B,D	A,B,D	A,B,C	A,D	B	C	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	4.24	1.00	0.11	1.00	4.00	9.00	0.75	3.57		

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)**PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A,B,C)**

Sol. $Ta_1 + \frac{T}{2}a_2 + \frac{T}{4}a_3 + \dots = 0$

$$\left(\frac{T}{m} - g\right) + \frac{1}{2}\left(\frac{T}{2m} - g\right) + \frac{1}{4}\left(\frac{T}{4m} - g\right) + \dots = 0$$

$$\frac{T}{m}\left(1 + \frac{1}{4} + \frac{1}{16} + \dots\right) = g\left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right)$$

$$T = \frac{3mg}{2}$$

$$\therefore a_1 = \frac{g}{2}$$

2. Ans. (B, D)**Sol.** Induced emf in the loops,

$$\varepsilon = \frac{R^2\theta}{2} \frac{dB}{dt} = \frac{a_0 R^2\theta}{2}$$

Induced current in the loop,

$$I = \frac{\varepsilon}{2R(1 + \cos\theta)\lambda} = \frac{a_0 R\theta}{4\lambda(1 + \cos\theta)}$$

$$V_o - V_q = I\lambda 2R \cos\theta = \frac{a_0 R^2\theta \cos\theta}{2(1 + \cos\theta)}$$

At $\theta = 45^\circ$,

$$V_o - V_q = \frac{a_0 R^2\pi}{8(\sqrt{2} + 1)}$$

$$V_P - V_Q = \varepsilon - I\lambda R = \frac{a_0 R^2\theta}{2} - \frac{a_0 R^2\theta}{4(1 + \cos\theta)}$$

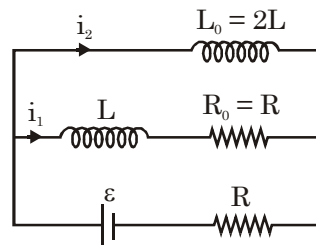
$$= \frac{a_0 R^2\theta}{4} \left(\frac{1 + 2\cos\theta}{1 + \cos\theta} \right)$$

For $\theta = 45^\circ$

$$V_P - V_Q = \frac{a_0 R^2\pi}{8\sqrt{2}}$$

3. Ans. (A, C)**Sol.** Current through the inductor ' L_0 ' in the steady state after closing the switch-2 will

$$\text{be } I_s = \frac{\varepsilon}{R}$$



$$\text{Now, } 2L \frac{di_2}{dt} = L \frac{di_1}{dt} + i_1 R_0$$

$$2L \int_0^{\varepsilon/R} di_2 - L \int_0^0 di_1 = R_0 \int_0^0 i_1 dt$$

$$2L \frac{\varepsilon}{R} - L \left(0 - \frac{\varepsilon}{2R} \right) = R_0 \Delta q$$

$$\frac{5L\varepsilon}{2R} = R_0 \Delta q \Rightarrow \Delta q = \frac{5L\varepsilon}{2RR_0}$$

$$\Delta q = \frac{5L\varepsilon}{2R^2}$$

4. Ans. (A, B, C, D)**Sol.** $X_C = -100\hat{j}$, $X_L = 200\hat{j}$

$$\frac{1}{z_{AB}} = \frac{1}{200\hat{j}} + \frac{1}{-100\hat{j}}$$

$$z_{AB} = -200\hat{j}$$

For circuit, $z = 100 - 200\hat{j}$

$$i_{OR} = \frac{V_0}{z} = \frac{100}{100(1 - 2\hat{j})} = \frac{1 + 2\hat{j}}{5}$$

$$i_{OR} = \frac{1}{\sqrt{5}} \sin(100t + \tan^{-1} 2)$$

$$V_R = 20\sqrt{5} \sin(100t + \tan^{-1} 2)$$

$$V_{AB} = \frac{1}{\sqrt{5}} \times (-200\hat{j}) = -40\sqrt{5}\hat{j}$$

$$V_L = V_C = 40\sqrt{5} \sin(100t + \tan^{-1} 2 - \frac{\pi}{2})$$

$$i_{OC} = \frac{V_C}{X_C} = \frac{40\sqrt{5}}{-100\hat{j}} = \frac{2}{\sqrt{5}}\hat{j}$$

$$i_C = \frac{2}{\sqrt{5}} \sin(100t + \tan^{-1} 2)$$

$$i_{OL} = \frac{40\sqrt{5}}{200j} = -\frac{1}{\sqrt{5}}j$$

$$i_L = \frac{1}{\sqrt{5}} \sin(100t + \tan^{-1}2 - \pi)$$

$$\text{At } t = 0, i_L = \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = -\frac{2}{5} \text{ A},$$

$$i_C = +\frac{2}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = \frac{4}{5}$$

$$V_C = V_L = -40\sqrt{5} \times \frac{1}{\sqrt{5}} = -40 \text{ Volt}$$

Power gained by inductor

$$= \left(-\frac{2}{5}\right) \times (-40) = 16 \text{ J/S}$$

$$\text{Power gained by capacitor} = \left(\frac{4}{5}\right) \times (-40)$$

$$\text{At } t = 0, i_R = \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = \frac{2}{5} \text{ A}$$

$$V_R = 20\sqrt{5} \times \frac{2}{\sqrt{5}} = 40 \text{ V}$$

$$\text{Power dissipated} = \frac{2}{5} \times 40 = 16 \text{ J/s}$$

5. Ans. (A,B,D)

6. Ans. (A,B,D)

7. Ans. (A)

8. Ans. (D)

9. Ans. (C)

10. Ans. (A)

SECTION-II

1. Ans. 47.77 to 47.78

2. Ans. 25.00

$$\text{Sol. } r = \frac{n^2 h^2}{\pi m z e^2} = \frac{n^2 h^2}{\pi \times 208 \times 3e^2} = \frac{h^2}{\pi m e^2}$$

$$\Rightarrow n \approx 25$$

3. Ans. 2.20

4. Ans. 1.84

5. Ans. 10.20

6. Ans. 3.00

7. Ans. 4.00

Sol. F = 5 Mk

$$F - 4Mk = 4Ma$$

$$a = \left(\frac{F}{20M}\right); V_1^2 = V^2 + 2\left(\frac{F}{20M}\right)(3200);$$

$$0 = V_1^2 + 2\left[-\frac{F}{5M}\right]S_1$$

$$S_1 = \frac{5MV^2}{2F} + \frac{5MF}{20Fm} \times 3200 + 3200;$$

$$S_1 = \frac{5MV^2}{2F} + 800 + 3200$$

$$0 = V^2 - \frac{2F}{5M}S_2; S_2 = \frac{5MV^2}{2F};$$

$$S_1 - S_2 = 4000 \text{ M} = 4.0 \text{ km}$$

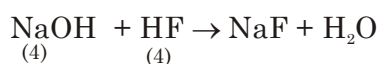
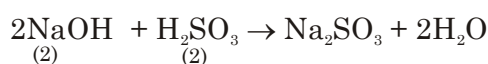
8. Ans. 12.90 to 13.00

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

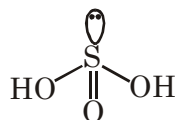
1. Ans. (B,C,D)



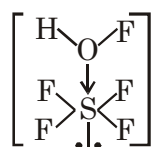
$\therefore$ 6 moles of NaOH is required
(6 = 0.5 $\times$ 12)



nature of π bond
 $2p\pi-3d\pi$



#



sp^3d^2 hybridised state
in the transition state

2. Ans. (A,B,C)

3. Ans. (A,B,D)

$$(N \times V)_{I_2} = (2 \times 1) \times 250 = 500$$

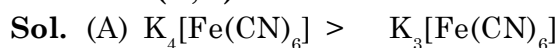
$$\text{n. f. for Cu}_2\text{S} = 10, N \times V = (0.5 \times 10) \times 100 = 500$$

$$\text{n. f. for CuS} = 8, N \times V = (0.5 \times 8) \times 125 = 500$$

$$\text{n. f. for S}_2\text{O}_3^{2-} = 1, N \times V = (0.5 \times 1) \times 1000 = 500$$

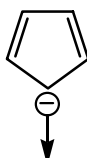
$$\text{n. f. for I}_2 = 2$$

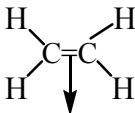
4. Ans. (A,B)

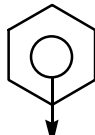


$$26 - 2 + 12 = 36 \quad 26 - 3 + 12 = 35$$

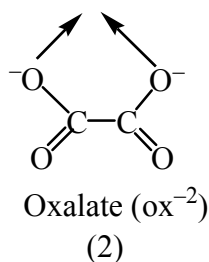
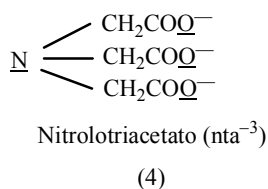
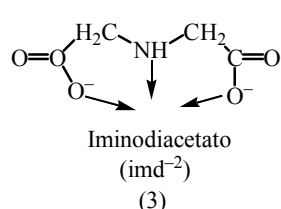
(B) As -ve charges on metal increases extent of synergic bonding increases, and CO bond order decreases.

(C) $\sigma\text{-C}_5\text{H}_5^-$  $\eta^1\text{-cyclopentadienyl}$

$\pi\text{-C}_2\text{H}_4$  $\eta^2\text{-ethylene}$

$\pi\text{-C}_6\text{H}_6$  $\eta^6\text{-benzene}$

(D)



5. Ans.(B,D)

6. Ans.(A,B,C)

7. Ans.(D)

Sol. (P) $\text{C} \Rightarrow \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \pi^* 2p_x^2 \approx \pi 2p_y^2$

B.O. = 2, (B.O.)_σ = 0, (B.O.)_π = 2

(Q) $\text{B}_2 \Rightarrow \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2p^2 < \pi 2p_x^1 \approx \pi 2p_y^1$

B.O. = 1, (B.O.)_σ = 0, (B.O.)_π = 1

(R) $\text{O}_2 \Rightarrow \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma^* 2p_z^2 < \pi 2p_x^2 \approx \pi 2p_y^2 < \pi^* 2p_x^1 \approx \pi^* 2p_y^1$

B.O. = 2, (B.O.)_σ = 1, (B.O.)_π = 1

(S) $\text{N}_2 \Rightarrow \sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 \approx \pi 2p_y^2 < \sigma 2p_z^2$

B.O. = 3, (B.O.)_σ = 1, (B.O.)_π = 2

If in molecule electron is removed from B.M.O. than ionisation energy of molecule is more than atom. If in molecule electron is removed ABMO from than ionisation energy of molecule is more than atom.

8. Ans.(C)

9. Ans.(A)

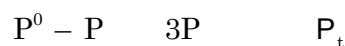
10. Ans.(D)

SECTION-II

1. Ans (3.00)

2. Ans.(0.03)

Sol. Reaction belongs to first order kinetics



$$3\text{P}^0 = 600 \Rightarrow \text{P}^0 = 200$$

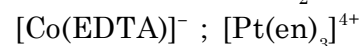
$$\text{P}^0 - \text{P} + 3\text{P} = \text{P}_t = 100$$

$$K_x = \frac{1}{70} \ln \frac{200}{200-100} = \frac{0.7}{70} = 0.01$$

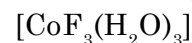
$$K_y = 3 \times K_x = 3 \times 0.01 = 0.03 \text{ min}^{-1}$$

3. Ans.(1.33)

Sol. Active [Be(bcac)₂], [Zn(gly)₂]



Inactive [CoBrCl(NO₂)₄]³⁻; [Pt(en)Cl₂]



$$\text{Ans.} = \frac{4}{3} \approx 1.33$$

4. Ans.(5.00)

II, III, IV, V, VI

5. (30.00)

KE = 0.9 eV per photoelectron

Total no. of photoelectrons

$$= \frac{1296}{0.9 \times 1.6 \times 10^{-19}} = 9 \times 10^{21}$$

$$9 \times 10^{21} = \int_0^t 10^{18} t^2 dt$$

$$9 \times 10^{21} = \frac{10^{18}}{3} \times t^3$$

$$t = 30 \text{ sec}$$

6. Ans.(83.00)

Sol. $97 \times 2 + 20 = 360$

$$\theta = 83.00$$

7. Ans. (3.00)

8. Ans (375.00)

For the given solution

$$\Delta T_f = 2 \times K_f \quad \dots (A)$$

560gm solution contains 1 mole urea & 500gm solvent. On placing solution at temperature 5°C below its freezing point then

$$(\Delta T_f + 5) = \frac{1000}{250} \times K_f = 4K_f \quad \dots (B)$$

from (A) & (B)

$$K_f = 5/2$$

$$\Delta T_f = 5$$

If the same solution is placed at temperature 15°C below its freezing then

$$\Delta T_f + 15 = K_f \times m$$

$$(5 + 15) = \frac{5}{2} \times \frac{1 \times 1000}{(\text{wt})_{\text{solvent}}}$$

$$(\text{Wt})_{\text{solvent}} = 125 \text{ gm}$$

$$\text{weight of solvent freezes out} = (500 - 125) = 375 \text{ gm}$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,B)

$$14 = 2^x + 4^x + 8^x$$

$$\Rightarrow x = 1$$

$$f'(x) = 2^x \ln 2 + 4^x \ln 4 + 8^x \ln 8$$

$$f'(1) = 2 \ln 2 + 8 \ln 2 + 24 \ln 2$$

$$f'(1) = 34 \ln 2$$

$$g'(14) = \frac{1}{34(\ln 2)}$$

2. Ans. (A,C)

$$\because \sqrt{y^2 + 2x} + \sqrt{y^2 - 2x} = 2x^2$$

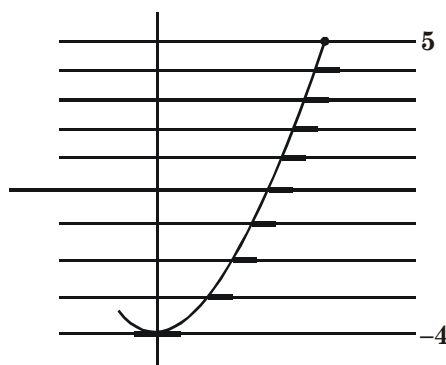
$$\therefore \sqrt{y^2 + 2x} - \sqrt{y^2 - 2x} = \frac{2}{x}$$

$$\Rightarrow y^2 + 2x = \left(x^2 + \frac{1}{x}\right)^2 \text{ or } y^2 = x^4 + \frac{1}{x^2}$$

$$\Rightarrow y = \sqrt{x^4 + \frac{1}{x^2}}$$

3. Ans. (B,D)

$$\text{Sol. } f(x) = [x^2 - 4] \forall x \in \left[\frac{-1}{3}, 3\right]$$



$$\Rightarrow x^2 - 4 \in [-4, 5]$$

discontinuous at 9 points

$$g(x) = [x^2 - 4](|x - 2| + |3x - 5|)$$

4. Ans. (A,B,D)

$$f(x) = (x - 1) \tan^{-1} x - \frac{1}{2} \ln(1 + x^2)$$

$$f'(x) = \tan^{-1} x - \frac{1}{1 + x^2}$$

$$f'(x) > 0 \forall x \geq 1$$

$\Rightarrow f(x)$ is increasing function

$$f''(x) > 0$$

$\Rightarrow f'(x)$ is increasing function $\forall x \geq 1$

$$\lim_{x \rightarrow \infty} f'(x) \rightarrow \frac{\pi}{2} \Rightarrow f'(x) < \frac{\pi}{2} \forall x \geq 1$$

Using LMVT in $[x, x + 4], x \geq 1$

$$\frac{f(x+4) - f(x)}{x+4-x} = f'(c) \text{ for } c \in (x, x+4)$$

$$\therefore f(x+4) - f(x) < 2\pi \quad \forall x \in [1, \infty)$$

5. **Ans. (A,B,C)**

$$F(x) = \int P(x) dx$$

$$\Rightarrow F(x) = x^9 - 3x^6 - 5x^4 + 15x + C$$

$$F(x) = x(x^5 - 5)(x^3 - 3) + C$$

Now use Rolle's theorem on $F(x)$ in

$$[0, 5^{1/5}], [5^{1/5}, 3^{1/3}] \text{ and } [0, 3^{1/3}]$$

6. **Ans. (A,D)**

Sol. $\int \sqrt{1 + \operatorname{cosec} x} dx = \int \frac{\cos x}{\sqrt{\sin x - \sin^2 x}} dx \dots (1)$

$$\text{Let } \sin x = t, \int \frac{dt}{\sqrt{t-t^2}} = \int \frac{dt}{\sqrt{\frac{1}{4} - \left(t - \frac{1}{2}\right)^2}}$$

$$= \sin^{-1}(2t-1) + c$$

$$= c - \sin^{-1}(1-2\sin x)$$

$$= c + \cos^{-1}(1-2\sin x)$$

OR

$$\text{Let } \sin x = t^2$$

from (1)

$$\int \frac{2t dt}{\sqrt{t^2 - t^4}} = 2 \int \frac{dt}{\sqrt{1-t^2}}$$

$$= 2 \sin^{-1} t + c = 2 \sin^{-1}(\sqrt{\sin x}) + c$$

7. **Ans. (B)**

$$f(x) = x^3 - 6x^2 - 36x + 2$$

$$f'(x) = 3x^2 - 12x - 36 = 3(x+2)(x-6)$$

$$f''(x) = 6x - 12 = 6(x-2)$$

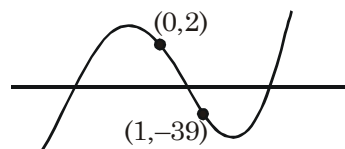
$$(P) \quad f'(x) > 0 \therefore x \in (-\infty, -2) \cup (6, \infty)$$

$$(Q) \quad f''(x) = 0 \therefore x = 2 \Rightarrow c = 2$$

$$(R) \quad f(x) = 0 \Rightarrow x^3 - 6x^2 - 36x + 2 = 0$$

$$\therefore f(0) = 2 \text{ and } f(1) = -39$$

$\Rightarrow$ three real roots



(S) for Rolle's theorem $f(a) = f(0)$

$$a^3 - 6a^2 - 36a + 2 = 2$$

$$\therefore a = 0, \frac{6 + \sqrt{180}}{2}, \frac{6 - \sqrt{180}}{2}$$

$$\therefore a > 2 \Rightarrow k = \frac{6 + \sqrt{180}}{2}$$

8. **Ans. (C)**

$$(P) \quad y = -2 \ln |x| \text{ as } |x| < 1$$

$$\Rightarrow y' = -\frac{2}{x} \Rightarrow y'\left(-\frac{1}{2}\right) = 4$$

$$(Q) \quad f(1) = \frac{1}{2} \Rightarrow g\left(\frac{1}{2}\right) = 1$$

$$g''(y) = -\frac{f''(x)}{(f'(x))^3}$$

$$g''\left(\frac{1}{2}\right) = -\frac{f''(1)}{(f'(1))^3}$$

$$\text{also } f' = \frac{3}{2}x^2 + \frac{1}{1+x^2}$$

$$f'' = 3x - \frac{2x}{(1+x^2)^2}$$

$$f'(1) = 2, f''(1) = 3 - \frac{1}{2} = \frac{5}{2}$$

$$g''\left(\frac{1}{2}\right) = -\frac{\frac{5}{2}}{(2)^3} = -\frac{5}{16}$$

$$(R) \quad \ell n y = x^x \ell n x$$

$$\frac{1}{y} y' = x^x (1 + \ln x) \ln x + \frac{x^x}{x} \Rightarrow y'(1) = 1$$

$$\Rightarrow \frac{yy'' - (y')^2}{y^2}$$

$$= x^x (1 + \ln x)^2 \ln x + x^x \left(\frac{1}{x} + \frac{2 \ln x}{x} \right) + \frac{x^x (1 + \ln x)}{x} - \frac{x^x}{x^2}$$

$$\text{at } x = 1 \Rightarrow y'' - 1 = 0 + 1 + 1 - 1 \Rightarrow y'' = 2$$

9. **Ans. (A)**

(P) $f'(x) = 0$ must have two real and unequal roots

$$\Rightarrow f'(x) = 6x^2 - 6(2 + \lambda)x + 12\lambda$$

$$D > 0 \Rightarrow (2 + \lambda)^2 - 8\lambda > 0 \Rightarrow (\lambda - 2)^2 > 0$$

(Q) Take log & differentiable

(R) Point of intersection is $x = 1$
now proceed

$$(S) \frac{dy}{dx} = 0 \text{ or } \infty$$

10. **Ans. (D)**

Case-I : If $a > 0$, then

$$g \circ f(x) = \begin{cases} x + a + 1; & x < -a \\ (x + a)^2 + 2(b - x - a) + 1; & -a \leq x < 0 \\ (1 - x)^2 + 2(b - 1 + x) + 1; & 0 \leq x < 1 \\ (x - 1)^2 + 2(b - x + 1) + 1; & x \geq 1 \end{cases}$$

$$a = 1 \text{ \& } b = 0$$

Case-II : If $a = 0$, then

$$g(f(x)) = \begin{cases} x + 1 & x < 0 \\ x^2 + 2b & 0 \leq x < 1 \\ (x - 2)^2 + 2b & x \geq 1 \end{cases}$$

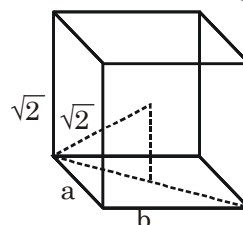
$$a = 0; b = \frac{1}{2}$$

$$\left. \frac{d(g(f(x)))}{dx} \right|_{x=4} = (2(x-1) - 2)|_{x=4} = 4$$

SECTION-II

1. **Ans. 4.24**

Let the other edges be 'a' & 'b'



$$\left(\frac{\sqrt{a^2 + b^2}}{2} \right)^2 + \left(\frac{\sqrt{2}}{2} \right)^2 = 2$$

$$\frac{a^2 + b^2}{4} = \frac{3}{2} \Rightarrow a^2 + b^2 = 6$$

Now, volume of cuboid = $ab\sqrt{2}$

$$\therefore \frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2}$$

$$\frac{a^2 + b^2}{2} \geq ab \Rightarrow ab \leq 3$$

$$\therefore V_{\max} = 3\sqrt{2}$$

2. **Ans. 1.00**

$$\text{Sol. } I = \int e^x \left(\frac{x^2 + 1}{(x^2 + 1)^{3/2}} - \frac{x}{(x^2 + 1)^{3/2}} \right) dx$$

$$\Rightarrow I = \int e^x \left(\frac{1}{\sqrt{x^2 + 1}} + \left(-\frac{x}{(x^2 + 1)^{3/2}} \right) \right) dx$$

$$I = \frac{e^x}{\sqrt{x^2 + 1}} + C$$

$$\therefore f(x) = \frac{1}{\sqrt{x^2 + 1}}$$

Range is (0, 1]

3. **Ans. 0.11**

$$x = 1, f(1) + 2f(1) = 1$$

$$\Rightarrow f(1) = \frac{1}{3}$$

$$f(x^2) + 2x^2f'(x^2) + (x^2 + 1)f'(x) + 2xf(x) = 1$$

$$x = 1$$

$$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$$

$$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$$

$$2xf'(x^2) + 4x^3f''(x^2) + 4xf'(x^2) + 2xf'(x) + (x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$$

$$x = 1$$

$$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1) + 2f(1) + 2f'(1) = 0$$

$$6f''(1) + \frac{2}{3} = 0$$

$$f''(1) = -\frac{1}{9}$$

4. Ans. 1.00

Sol. $\ell_1 = (x \cos x)^x \quad (0^0)$

$$\log \ell_1 = \lim_{x \rightarrow 0^+} \frac{\ell \ln x \cos x}{1/x}$$

$$= \frac{1}{x \cos x} (\cos x - x \sin x)$$

$$= \frac{-1/x^2}{-1/x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-x(\cos x - x \sin x)}{\cos x} = 0$$

$$\ell_1 = e^0 = 0$$

$$\Rightarrow \ell_2 = \lim_{x \rightarrow 0^+} (x \sin x)^{1/x} = 0^{+\infty} = 0$$

$$\ell_1 + \ell_2 = 1$$

5. Ans. 4.00

$$x = 3 \cos \theta - \cos^3 \theta \Rightarrow \frac{dx}{d\theta} = -3 \sin^3 \theta$$

$$y = 3 \sin \theta - \sin^3 \theta \Rightarrow \frac{dy}{d\theta} = 3 \cos^3 \theta$$

$$\therefore \frac{dy}{dx} = -\frac{\cos^3 \theta}{\sin^3 \theta}$$

equation of normal is

$$y - (3 \sin \alpha - \sin^3 \alpha) = \frac{\sin^3 \alpha}{\cos^3 \alpha} (x - (3 \cos \alpha - \cos^3 \alpha))$$

$$\Rightarrow y \cos^3 \alpha - x \sin^3 \alpha = \frac{3}{4} \sin 4\alpha$$

$$\Rightarrow 4 \left[\frac{\cos^4 \alpha}{\cos \alpha} \cdot y - \frac{\sin^4 \alpha}{\sin \alpha} \cdot x \right] = 3 \sin 4\alpha$$

$$\therefore a = \cos^4 \alpha, b = \sin^4 \alpha, a + b = \sin 4\alpha$$

$$\Rightarrow \cos^4 \alpha + \sin^4 \alpha = \sin 4\alpha$$

$$\Rightarrow 2 - \sin^2 2\alpha = 2 \sin 4\alpha$$

$$\Rightarrow 4 - [1 - \cos 4\alpha] = 4 \sin 4\alpha$$

$$\Rightarrow 3 = -\cos 4\alpha + 4 \sin 4\alpha$$

$$\Rightarrow \text{Total 4 values of } \alpha$$

6. Ans. 9.00

$f(x)$ is discontinuous at all integers except at $x = 0$

7. Ans. 0.75

$$\therefore 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow 2.4 \frac{dx}{dt} + 2.3(-1) = 0 \Rightarrow \frac{dx}{dt} = \frac{3}{4}$$

8. Ans. 3.57

$$\therefore y = \frac{2x^2 + 7x + 3}{2x^2 + x + 1} \quad \dots(i)$$

$$\Rightarrow 2x^2y + xy + y = 2x^2 + 7x + 3$$

$$\Rightarrow 2x^2(y - 1) + x(y - 7) + (y - 3) = 0$$

$$\text{if } y - 1 \neq 0$$

$$\therefore x \in \mathbb{R} \Rightarrow D \geq 0$$

$$(y - 7)^2 - 8(y - 1)(y - 3) \geq 0$$

$$\Rightarrow y^2 - 14y + 49 - 8y^2 + 32y - 24 \geq 0$$

$$\Rightarrow -7y^2 + 18y + 25 \geq 0$$

$$\Rightarrow 7y^2 - 18y - 25 \leq 0$$

$$\Rightarrow (y + 1)(7y - 25) \leq 0$$

$$-1 \leq y \leq \frac{25}{7} \quad \text{but } y \neq 1$$

$$\text{put } y = 1 \text{ in (1) } \therefore (1)$$

$$\Rightarrow 2x^2 + 7x + 3 = 2x^2 + x + 1$$

$$\Rightarrow x = -\frac{1}{3}$$